BACHELOR OF COMPUTER APPLICATIONS

SEMESTER 1

FUNDAMENTALS OF MATHEMATICS

Unit 7

Special Functions Part 2

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1. INTRODUCTION

Unit 6 delves deep into the core of trigonometry, starting from the basics of positive and negative angles, embracing the radian measure, and elegantly transitioning to the definition of trigonometric functions via the unit circle. We explore not just the domain and range of these cyclic functions but also their intricate graphs, offering a vivid portrayal of their periodic nature. Trigonometric equations, with their general solutions, unfold the symphony of sines, cosines, and tangents in harmonizing the values across different quadrants.

The realm then expands to the inverse trigonometric functions, where we step beyond the basic functions to their inverses, exploring their definitions, domains, ranges, and principal value branches. The graphical representation of these functions not only provides a visual understanding but also sheds light on their elementary properties, adding another dimension to our trigonometric knowledge.

Expanding our horizon, we venture into exponential functions, exploring their essence, the compelling properties that govern their growth and decay, and the calculus that lies beneath - their derivatives. This journey through Unit 6 not only equips us with a profound understanding of these mathematical tools but also prepares us for their versatile applications in various fields.

As we transition to Unit 7, we gear up to embrace the logarithmic functions, the counterparts of exponentials, unravelling their unique properties and how they simplify complex mathematical expressions. The focus on logarithmic differentiation showcases the elegance of calculus in handling functions involving products, quotients, and powers, further enriched by exploring functions expressed in parametric forms and the intriguing world of second-order derivatives.

Hyperbolic functions await, introducing us to the 'hyperbolic trigonometry' analogous to circular functions but associated with hyperbolas. We explore their definitions, properties, and the remarkable identities they satisfy, akin to their trigonometric counterparts. The introduction to inverse hyperbolic functions adds another layer to our mathematical palette, enhancing our problem-solving toolkit.

Approaching these units, a blend of theoretical understanding and practical application is paramount. Visualizing graphs, solving equations, and applying properties should go together with understanding the theoretical underpinnings of each function and its derivative. Embracing a variety of problems, from the simple to complex, ensures a robust grasp of the concepts.

1.1 Learning Objectives

At the end of this unit, you will be able to:

- *Identify the definitions of basic trigonometric functions and inverse trigonometric functions.*
- *Explain the relationship between exponential functions and their logarithmic counterparts.*
- *Calculate the values of trigonometric functions for given angles in both degrees and radians.*
- *Differentiate between the graphs of exponential functions and their respective logarithmic functions.*

2. LOGARITHMIC FUNCTIONS

Logarithmic functions are an integral part of mathematics, providing a powerful tool for solving problems involving exponential relationships. Their unique properties and applications make them indispensable in fields such as physics, engineering, and economics.

A logarithmic function is the inverse of an exponential function and is defined as follows:

$$y = log_n(x)$$

where,

- x is the argument of the logarithmic function
- a is the base of the logarithm, which is a positive real number not equal to 1
- y is the exponent to which the base a must be raised to produce x

This can be equivalently expressed as

$$a^y = x$$

The most encountered logarithms are those with base 10, known as common logarithms, denoted as log(x), and those with base e, known as natural logarithms, denoted as ln(x).

Properties of logarithmic function

Logarithmic functions exhibit several important properties that facilitate the manipulation and simplification of logarithmic expressions:

1. Product Property: The logarithm of a product is the sum of the logarithms of the factors.

$$log_a(xy) = log_a x + log_a y$$

2. Quotient Property: The logarithm of a quotient is the difference between the logarithm of the numerator and the logarithm of the denominator.

$$log_a\left(\frac{x}{y}\right) = log_a x - log_a y$$

3. Power Property: The logarithm of a power is the exponent times the logarithm of the base.

$$log_a(x^n) = n \cdot log_a x$$

4. Change of Base Formula: Allows the expression of a logarithm in one base in terms of logarithms in another base.

$$log_a x = \frac{log_b x}{log_b a}$$

5. Inverse Property: The exponential function and the logarithmic function are inverses of each other.

$$a^{log_a x} = x$$

$$log_a a^x = x$$

Derivatives of Logarithmic Functions

The derivatives of logarithmic functions are particularly useful in calculus, as they enable the calculation of the rate of change of logarithmic expressions.

Derivative of the Natural Logarithm:

$$\frac{d}{dx} log_n x = \frac{1}{x}$$

This formula states that the derivative of the natural logarithm of x is $\frac{1}{x}$, applicable for $x > 0$.

Derivative of the Logarithm with Base a:

$$\frac{d}{dx} log_a x = \frac{1}{x\, log_n a}$$

This shows that the derivative of $log_a x$ involves the natural logarithm of the base a, highlighting the intrinsic relationship between different logarithmic bases and the natural logarithm.

These derivative formulas are fundamental in solving various problems in calculus, especially those involving growth and decay modelled by logarithmic functions.

Logarithmic functions, with their distinctive properties and applications, play a crucial role in mathematics and its applications. Understanding their definition, key properties, and how to differentiate them opens up a wide array of possibilities for solving equations, simplifying expressions, and modelling real-world phenomena. The interplay between exponential and logarithmic functions, particularly evident in their inverse relationship and derivative formulas, underscores the coherence and depth of mathematical concepts.

Example:

Solve for x in the equation $log_2(x) + log_a(x-2) = 3$

Solution:

First, use the product property of logarithms to combine the logarithms on the left side.

$$log_2\big(x(x-2)\big) = 3$$
$$log_2(x^2 - 2x) = 3$$

Using the definition of a logarithm, convert this equation to its exponential form.

$$2^3 = x^2 - 2x$$
$$8 = x^2 - 2x$$
$$x^2 - 2x - 8 = 0$$

This is a quadratic equation, which can be solved using the quadratic formula. The roots are:

$$x = \frac{-(-2) \pm \sqrt{(-2)^2 - 4 \cdot 1 \cdot (-8)}}{2 \cdot 1}$$
$$x = 1 \pm \sqrt{9}$$
$$x = 1 \pm 3$$

Therefore, the solutions are x = 4 and x = -2. However, since $log_2(x-2)$ must be defined, $x - 2 > 0$, thus $x > 2$. Hence, the only valid solution is x = 4.

Example 3:

Find the derivative of $f(x) = log_n(x^2 + 1)$

Solution:

Use the chain rule for differentiation, which states that if $f(x) = g\big(h(x)\big)$, then $f'(x) = g'\big(h(x)\big) \cdot h'(x)$

The derivative of $g(h) = log_n(h)$ with respect to h is $\frac{1}{h}$. So, $g'\big(h(x)\big) = \frac{1}{x^2+1}$

The derivative of $h(x) = x^2 + 1$ with respect to x is $2x$

Using the chain rule:

$$f'(x) = \frac{1}{x^2 + 1} \cdot 2x$$

Therefore,

$$f'(x) = \frac{2x}{x^2 + 1}$$

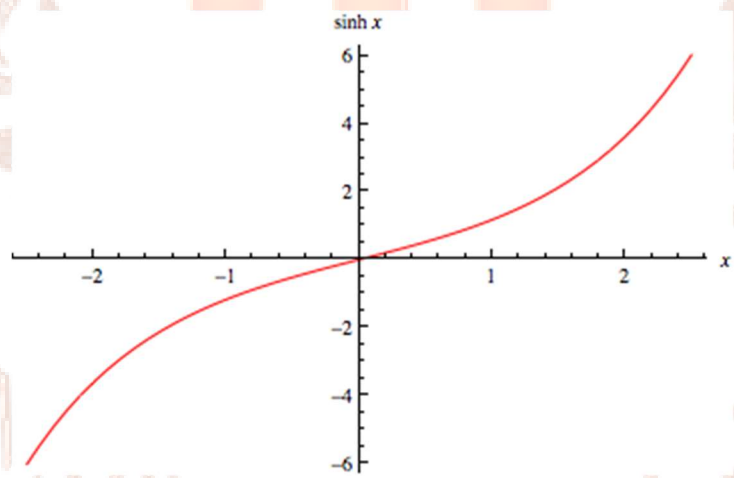
This derivative represents the rate of change of f(x) with respect to x.

3. HYPERBOLIC FUNCTIONS

Hyperbolic functions, often referred to as "hyperbolic trigonometric functions," are analogues of the circular trigonometric functions, but they relate to hyperbolas rather than circles. These functions are essential in various areas of mathematics, including calculus, complex analysis, and differential equations, as well as in physics, particularly in the theory of special relativity and wave propagation.

Hyperbolic Sine Function (sinh)

$$sinh(x) = \frac{e^x - e^{-x}}{2}$$

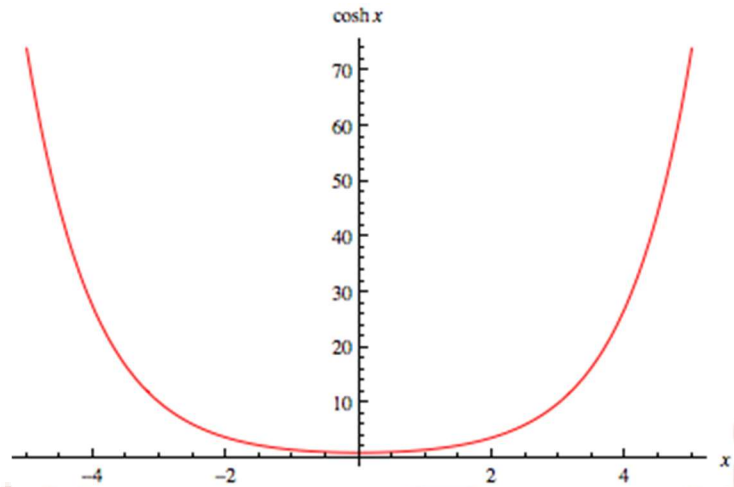


Hyperbolic Sine Function

The graph of sinh(x) is an odd function and resembles an S-shape, passing through the origin (0, 0). It exhibits unbounded growth as x increases or decreases.

Hyperbolic Cosine Function (cosh)

$$cosh(x) = \frac{e^x + e^{-x}}{2}$$

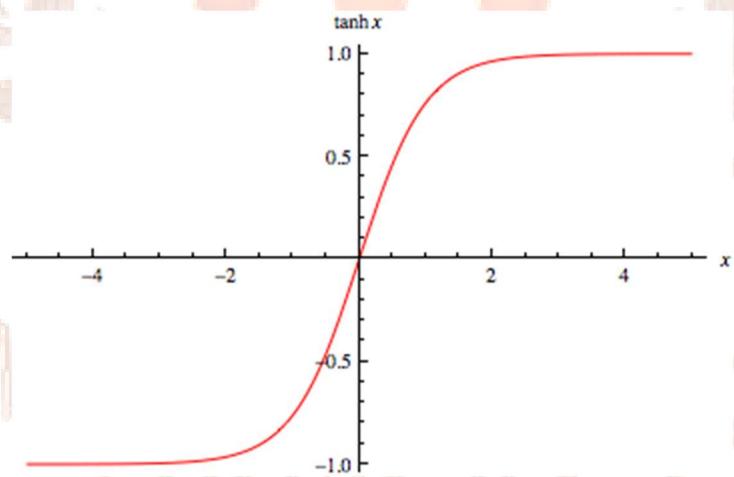


Hyperbolic Cosine Function

The graph of cosh(x) is an even function, with a minimum value of 1 at x=0. It resembles the shape of a hanging cable or chain, known as a catenary, and grows without bound as |x| increases.

Hyperbolic Tangent Function (tanh)

$$tanh(x) = \frac{sinh(x)}{cosh(x)} = \frac{e^x - e^{-x}}{e^x + e^{-x}}$$



Hyperbolic Tangent Function

The graph of tanh(x) is an odd function, with horizontal asymptotes at y=1 and y=−1 as x tends to infinity and negative infinity, respectively. It passes through the origin and is bounded between -1 and 1.

Properties of Hyperbolic Functions

Symmetry

$$sinh(-x) = -sinh(x)$$
$$cosh(-x) = cosh(x)$$

making, sinh(x) an odd function and cosh(x) an even function.

Unbounded Growth

cosh(x) and sinh(x) grow exponentially as x increases or decreases, but tanh(x) remains bounded between -1 and 1.

Derivatives

The derivatives of hyperbolic functions mirror those of their trigonometric counterparts, with

$$\frac{d}{dx}(sinh(x)) = cosh(x)$$

$$\frac{d}{dx}(cosh(x)) = sinh(x)$$

$$\frac{d}{dx}(tanh(x)) = sech^2(x)$$

Hyperbolic Functions Identities

Hyperbolic functions satisfy many identities similar to trigonometric identities, including:

Addition Formulas:

$$sinh(x+y) = sinh(x)\,cosh(y) + cosh(x)\,sinh(y)$$

$$cosh(x+y) = cosh(x)\,cosh(y) + sinh(x)\,sinh(y)$$

Double Angle Formulas:

$$sinh(2x) = 2\,sin\,h(x)\,cosh(x)$$

$$cosh(2x) = cosh^2(x) + sinh^2(x)$$

Hyperbolic Pythagorean Identity:

$$cosh^2(x) - sinh^2(x) = 1$$

Inverse Hyperbolic Functions

Inverse hyperbolic functions allow the determination of the hyperbolic angle corresponding to a given hyperbolic function value. They are defined as:

Inverse Hyperbolic Sine (arsinh or $\sinh^{-1}$)

$$sinh^{-1}(x) = log_n\left(x\sqrt{x^2+1}\right)$$

Inverse Hyperbolic Cosine (arcosh or $\cosh^{-1}$)

For $x \geq 1$,

$$cosh^{-1}(x) = log_n\left(x+\sqrt{x^2-1}\right)$$

Inverse Hyperbolic Tangent (artanh or $\tanh^{-1}$)

For $|x| < 1$,

$$tanh -1\,(x) = \frac{1}{2}(log_n(1+x) - log_n(1-x))$$

These inverse functions are crucial for solving equations involving hyperbolic functions and are analogous to the arcsin, arccos, and arctan functions for trigonometry.

Example:

Evaluate $sinh(0)\,, cosh(0)\,, tanh(0)$

Solution:

$$sinh(0) = \frac{e^0 - e^{-0}}{2} = \frac{1-1}{2} = 0$$

$$cosh(0) = \frac{e^0 + e^{-0}}{2} = \frac{1+1}{2} = 1$$

$$tanh(0) = \frac{sinh(0)}{cosh(0)} = \frac{0}{1} = 0$$

Example:

Find the derivative of $f(x) = sinh(x) + cosh(x)$

Solution:

$$f'(x) = \frac{d}{dx}f(x)$$

$$f'(x) = \frac{d}{dx}(sinh(x) + cosh(x))$$

$$f'(x) = \frac{d}{dx}sinh(x) + \frac{d}{dx}\cosh(x)$$

$$f'(x) = cosh(x) + sinh(x)$$

Hyperbolic functions are widely used in various branches of mathematics and physics. They are particularly significant in the study of hyperbolic geometry, special relativity, and the solutions of certain differential equations. For example, the shape of a hanging cable or chain, known as a catenary, is described by the hyperbolic cosine function. In special relativity, hyperbolic functions are used to describe the relationship between time dilation and velocity.

In engineering, hyperbolic functions are applied in areas such as signal processing, where they help to model hyperbolic responses in control systems and circuits.

Hyperbolic functions are a fascinating and powerful set of functions that extend the utility of traditional trigonometry to hyperbolas, much like how circular functions relate to the circle. Their unique properties, identities, and the ability to define inverse functions make them an invaluable tool in both theoretical and applied mathematics. Understanding the behaviour of hyperbolic functions, their graphical representations, and their applications in real-world scenarios underscores their importance across various scientific disciplines.

4. SUMMARY

Alright, let's wrap up what we've covered in this unit. It's been quite a journey through the mathematical landscapes of trigonometry, exponential functions, and those intriguing hyperbolic functions. Think of trigonometry as your go-to toolkit for anything that involves angles and circles. We started with the basics, exploring how to measure angles not just in degrees but in radians too – a concept that might have seemed a bit alien at first but is super handy, especially in calculus.

It's fascinating how a simple circle can unlock so much understanding of angles and their relationships. And it's not just about solving triangles in geometry class; these functions are everywhere, from the waves in the ocean to the oscillations in your phone's signal.

Then we dove into the world of inverse trigonometric functions. It's like we flipped the script on trig functions, asking not "what's the sine of this angle?" but "what angle gives me this sine?". This opened up a whole new perspective, didn't it? Especially when we plotted them, revealing those distinctive curves that mirror our original trig functions but with a twist.

If trig functions are the masters of cycles and oscillations, exponential functions are all about growth and decay. They show up in the most unexpected places, from calculating compound interest to understanding how populations expand or how radioactive materials break down. The exponential function, with its base 'e', is like the secret sauce in calculus, making differentiation and integration a breeze.

We couldn't leave out logarithms, the other side of the exponential coin. Logarithmic functions, with their unique properties, help us unravel exponential relationships, making complex multiplications and divisions a piece of cake. And when we explored derivatives of these functions, we saw the elegance of mathematics come to life, showing us how change at one moment relates to the function as a whole.

Hyperbolic functions were the cherry on top. These functions, resembling our trigonometric friends but related to hyperbolas instead of circles, have a beauty of their own. From sinh to cosh and tanh, they extend the realm of functions we can explore and understand, proving that mathematics is a never-ending adventure.

So, as we close this unit, reflect on how these concepts interconnect, weaving a rich tapestry of mathematical ideas that stretch from the purest of theories to the most practical applications in our daily lives. It's not just about solving equations; it's about seeing the world in a whole new light.

5. SELF-ASSESSMENT QUESTIONS

SELF-ASSESSMENT QUESTIONS – 1

1. What is the radian measure equivalent to 180°?
   a) π/2
   b) π
   c) 2π
   d) π/4
2. What is the value of sin(π/6)?
   a) 1/2
   b) √2/2
   c) √3/2
   d) 1
3. The inverse function of $y = log_b(x)$ is:
   a) $y = b^x$
   b) $y = x^b$
   c) $y = \frac{1}{b^x}$
   d) $y = \frac{1}{x^b}$
4. What is the derivative of $f(x) = e^x$?
   a) e^x
   b) xe^{x-1}
   c) x^e
   d) e^{x^2}

5. The hyperbolic cosine of 0, cosh(0), is:
   a) 0
   b) 1
   c) e
   d) -1
6. The range of the function $f(x) = sin^{-1}(x)$ is
   a) $[0, \pi/2]$
   b) $[-\pi/2, \pi/2]$
   c) $[0, \pi]$
   d) $[-\pi, \pi]$
7. What is $log_{10}(100)$
   a) 10
   b) 2
   c) 100
   d) 1
8. The derivative of $f(x) = log_n(x)$
   a) $1/x$
   b) $log_n(x)$
   c) x
   d) $\frac{x}{log_n(x)}$
9. Convert 270° to radians.
10. Find cos(π/4).
11. Solve for x if sin(x)=1/2 and 0≤x≤2π.
12. Evaluate $tan^{-1}(1)$
13. Calculate $\frac{d}{dx}(3^x)$
14. Find $cosh^{-1}(1)$
15. If $y = log_2(x)$,find $\frac{dy}{dx}$

16. Solve for x: $e^{2x} = 7$
17. Calculate $sin\,h(0)$
18. If f(x)=sin(x)cos(x), find $f'(x)$
19. Determine $\lim_{x\to 0} \frac{sin(x)}{x}$
20. Solve for y if $y = 2^x$ and x=3.

6. TERMINAL QUESTIONS

1. Convert 135° to radians.
2. Find the exact value of $sin\frac{2\pi}{3}$
3. Solve for x in the equation $cos(x) = \frac{-\sqrt{2}}{2}$ where 0≤x<2π.
4. Determine $tan^{-1}(\sqrt{3})$
5. If $f(x) = e^{2x}$, find $f'(x)$
6. Calculate $sinh^{-1}(1)$
7. Derive y with respect to x if $y = log_n(x^2 + 1)$
8. Solve for x in the exponential equation $4^x = 16$
9. Find cosh(2).
10. If g(x)=cos(x)sin(x), find $g'(x)$
11. Evaluate $\lim_{x\to 0} \frac{tan\,x}{x}$
12. Solve $log_5(x) + log_5(x - 1) = 1$
13. Convert $\frac{7\pi}{6}$ radians to degrees.
14. Find the derivative of $f(x) = 5^{x^2}$
15. Calculate $tanh^{-1}(1 / 2)$
16. Solve for y in $y = 10^{2x}$ when x = -0.5
17. Determine $sec^{-1}(2)$
18. If h(x)=ln(sin(x)), find $h'(x)$
19. Find the exact value of $cos\frac{5\pi}{4}$
20. Solve for x in the hyperbolic equation sinh(x)=cosh(x).

8. ANSWERS

Self-Assessment Questions

1. B) π
2. A) ½
3. A) $y = b^x$
4. A) e^x
5. B) 1
6. B) $[-{}^{\pi}/_{2}, {}^{\pi}/_{2}]$
7. B) 2
8. A) $1/x$
9. $\frac{3\pi}{2}$
10. $\frac{\sqrt{2}}{2}$
11. $x = \frac{\pi}{6}, \frac{5\pi}{6}$
12. $\frac{\pi}{4}$
13. $3^x \, log_n(3)$
14. 0
15. $\frac{1}{x \, log_n(2)}$
16. $log_n \frac{7}{2}$
17. 0
18. $cos\, 2x - sin\, 2x$
19. 1
20. $y = 8$

Terminal Questions

1. $\frac{3\pi}{4}$
2. $\frac{\sqrt{3}}{2}$
3. $x = \frac{3\pi}{4}, \frac{5\pi}{4}$
4. $\frac{\pi}{3}$
5. $2e^{2x}$

6. $\log_n(1+\sqrt{2})$
7. $\frac{2x}{x^2+1}$
8. $x = 2$
9. $\frac{e^4+e^{-4}}{2}$
10. $\cos(2x)$
11. 1
12. $x = 3$
13. 210^0
14. $2x\log_n(5)\cdot 5^{x^2}$
15. $\frac{1}{2}log_n(3)$
16. $y = 0\cdot 1$
17. $\frac{\pi}{3}, \frac{2\pi}{3}$
18. $cot(x)$
19. $-\frac{\sqrt{2}}{2}$
20. x = 0, (Since sinh(x) = cosh(x) implies $e^x - e^{\{-x\}} = e^x + e^{\{-x\}}$, which is not possible for any real x except 0)

9. REFERENCES

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